Sit/Stand/Walk/Stairs Controller for the 2nd Generation Open-Source Leg

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Version 1.0

This code release contains the main python scripts and a compiled library for running the Locolab's hybrid kinematic impedance controller (HKIC) on the second generation Open-Source Leg (OSL). This controller calculates the desired joint torques for the knee and ankle based on data-driven models and a thigh-based phase variable. It supports sit/stand, walking, and stair ambulation. This main script handles reading the sensors on the OSL, commanding joint torques, and logging data.

The compiled controller implements the control logic detailed in the following publications:

- T. K. Best, C. G. Welker, E. J. Rouse and R. D. Gregg, "Data-Driven Variable Impedance Control of a Powered Knee–Ankle Prosthesis for Adaptive Speed and Incline Walking," in *IEEE Transactions on Robotics*, vol. 39, no. 3, pp. 2151-2169, June 2023, doi: 10.1109/TRO.2022.3226887.
- C. G. Welker, T. K. Best and R. D. Gregg, "Improving Sit/Stand Loading Symmetry and Timing Through Unified Variable Impedance Control of a Powered Knee-Ankle Prosthesis," in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, doi: 10.1109/TNSRE.2023.3320692.
- R. J. Cortino, T. K. Best and R. D. Gregg, "Data-Driven Phase-Based Control of a Powered Knee-Ankle Prosthesis for Variable-Incline Stair Ascent and Descent," in IEEE Transactions on Medical Robotics and Bionics, vol. 6, no. 1, pp. 175-188, Feb. 2024, doi: 10.1109/TMRB.2023.3328656.
- T. K. Best, G. C. Thomas, S. R. Ayyappan, R. D. Gregg and E. J. Rouse, "A Compensated Open-Loop Impedance Controller Evaluated on the Second-Generation Open-Source Leg Prosthesis," in IEEE/ASME Transactions on Mechatronics, doi: 10.1109/TMECH.2024.3508469.

We kindly request that you cite the relevant publications in any research projects that use this control code.

License

This library is released under the Polyform Strict License.

Using The Library

Prerequisites:

1. Place the python files and the compiled library in the same directory[1] on the OSL's raspberry pi.
2. Ensure that all dependencies are installed. Create a virtual environment and install the requirements:

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

3. Ensure all required sensors are working. The controller requires knee and ankle actuators, joint encoders, a load cell, and a thigh IMU. See the OSL Library documentation for getting started with these sensors if you're unfamiliar.
4. Connect to the raspberry pi via ssh. We typically use Visual Studio Code as recommended by the Open-Source Leg Community.
5. Our script assumes that the knee actuator is powered on before the ankle actuator. Be sure you've powered the actuators on in the correct order before starting the main script.

Conventions:

Our code assumes the following conventions:

- Joint angles and torques are defined using standard biomechanics frames. Knee flexion and ankle dorsiflexion are considered positive.
- The global thigh angle is measured relative to gravity, with hip flexion corresponding to positive thigh angles. For the thigh IMU, the X direction should be normal to the sagittal plane.
- For the load cell, the Z axis should point axially along the shank. For this release, the only signal that matters is the Z force, and the sign does not matter.

[1] If you would rather keep the compiled library in a different location, you must update the `CONTROL_LIBRARY_PATH` parameter.

Basic Usage:

1. Start by running the main python script. The system will zero the loadcell and home the joints. Make sure the foot is off the ground and the shank is pointing straight up and down when zeroing the loadcell.
2. Then, you'll be prompted to begin walking. After pressing the "enter" key, the main control loop is entered.
3. To end the main loop, press ctrl+c in the terminal.

Configuration

There is a configuration block at the top of the main python script that allows you to change the controller settings. We explain each setting below:

Ambulation Modes

The controller allows two modes for classification: manual or automatic. The automatic mode implements the logic described in the Welker et al. 2023 to switch between sit/stand and walk. In this release, automatic classification for stairs is not enabled due to missing environmental sensors. In manual mode, you can select between walk, sit/stand, stairs, or peg. Peg mode controls to a nominal straight leg configuration for fittings or other non-steady activities if needed. Set the mode (the `eClassificationMode` enum) in the configuration block to choose between auto and manual. Set a value of `eControlMode` to choose between Peg, Sit/Stand, Stair, or Walk when using manual classification.

User Mass

The controller needs to know the user's mass to appropriately scale the impedance models. Set this value in kilograms in the configuration block.

Loadcell Calibration Matrix

You must replace the loadcell calibration matrix with the calibration matrix for your specific hardware. See your loadcell datasheet for details. We place our loadcell between the knee and ankle assemblies.

Task Variables

The walk and stair modes take task variable inputs to handle different terrains. For walk mode, set the walking speed (between 0.6 and 1.4 m/s) and ground incline (between -10 and 10 deg). For stair mode, set the ascent flag (true for going up stairs, false for going down stairs) and the stair incline (between 19 and 30 deg).

Friction Compensation

The `ENABLE_FRICTION_COMPENSATION` flag turns on or off friction compensation. Details for the friction compensation are in the TMECH paper listed above.

Using a Different Method for Global Thigh Angle Measurement

You may use another means to measure the global thigh angle if you would rather. For example, some research groups already have an Xsens system, and therefore prefer to use it instead of the Microstrain IMU. Note that the controller's behavior depends greatly on this signal, and it is very important to provide high-quality measurements of the thigh angle and velocity. Noisy or drifting values may cause undesired behaviors.

To use a different sensor, simply replace the writes to `hkic.inputs.hardware.sensors.thighIMU` with the appropriate readings from your sensor within the main loop.

Contact

Questions regarding the control release should be posted on the Open-Source Leg Forum. Posting on the forum allows the community to benefit from our discussions. For other inquiries regarding our research, please contact Dr. Robert Gregg at rdgregg@umich.edu.

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